PRISM | Tech Management, SRI-B

CLASH OF THE CLAWS

MSRIT_SDS
“See the world. Hear what matters”

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Problem Statement

Problem:

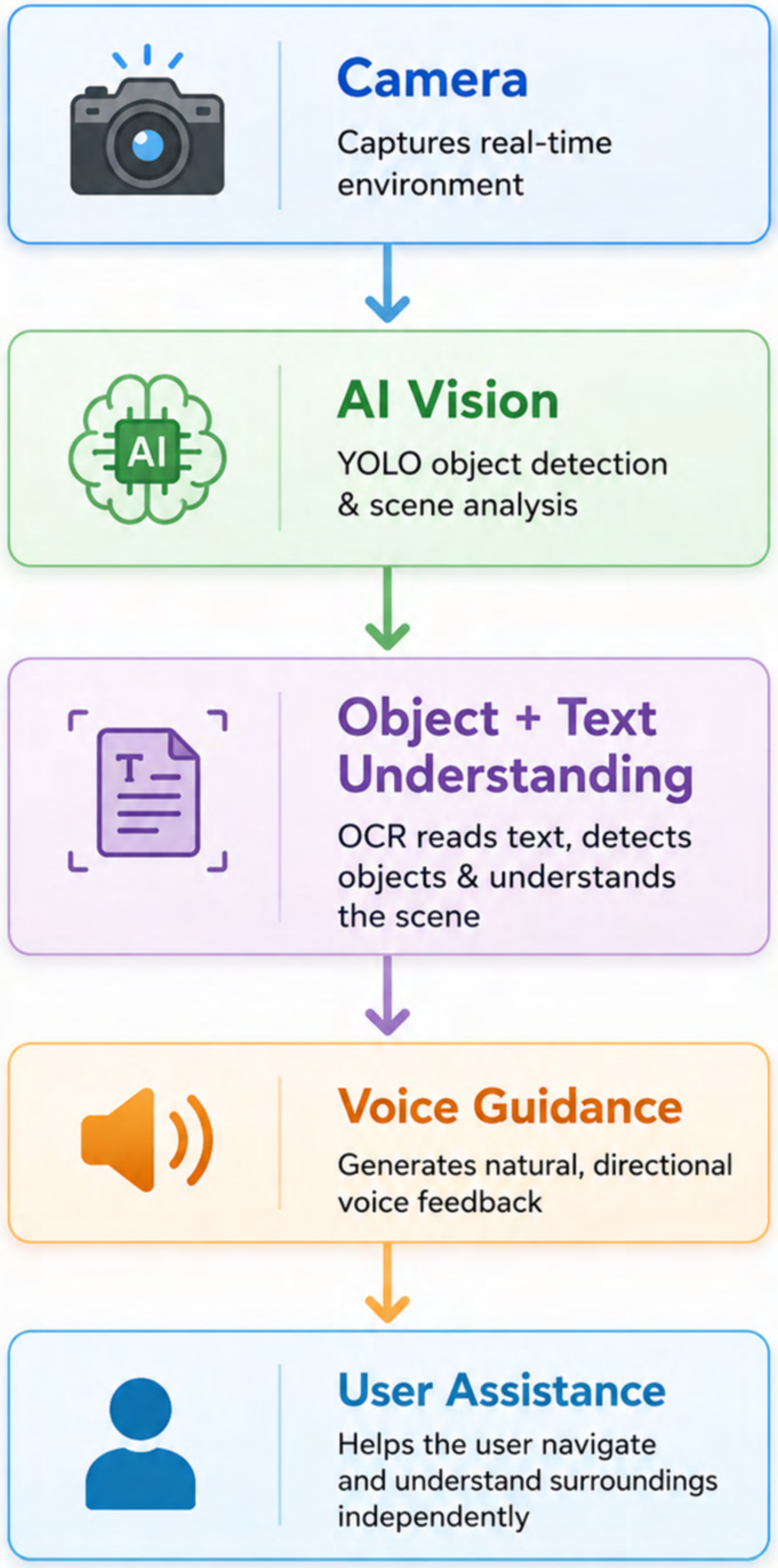
- Challenges Faced by Visually Impaired Users
- Difficulty understanding surroundings in real time
- Cannot easily read signs/screens/labels
- Lack of directional awareness
- Existing solutions depend on internet/cloud systems

Our Solution:

VisionLink

- Real-time offline assistive AI system that:
- Detects objects
- Reads environmental text
- Understands surroundings
- Provides directional voice guidance

“Helping visually impaired users perceive and navigate the world independently.”



Existing Solutions :

MICROSOFT SOUNDSCAPE

Audio-based outdoor navigation using GPS

Limitations:
No real-time obstacle detection (no camera, no precision)

NAVCOG

Indoor navigation using Bluetooth beacons

Limitations:
Requires pre-installed infrastructure (not usable everywhere)

MIT FINGERREADER

Helps read text with finger tracking

Limitations:
Limited to reading tasks, not full navigation

ORCAM MYEYE / GOOGLE LOOKOUT

Reads text and identifies objects using camera

Limitations:
No guidance on how to reach or interact with objects

AIRA

Human-assisted real-time guidance via video

Limitations:
Expensive, requires internet, not scalable

FingerReader

aira

NavCog
Indoor Navigation

Microsoft Soundscape

ORCAM MyEye

Google Lookout

VisionLink Solution

"An offline real-time assistive AI system for visually impaired users."

Core Features:

📷 REAL-TIME ENVIRONMENT UNDERSTANDING

- Detects nearby objects instantly
- Understands object positions (left/right/center)

📝 INTELLIGENT OCR READING

- Reads signs, screens, labels, notices
- Converts environmental text into speech

🎤 NATURAL VOICE INTERACTION

- Supports offline voice commands
- Hands-free user interaction using Vosk

🎧 SPATIAL AUDIO GUIDANCE

- Direction-aware audio feedback
- Helps users locate objects naturally

Key advantages:

✅ Offline-first
✅ Real-time
✅ OpenClaw AI
✅ Cross-platform
✅ Assistive AI

Idea

Demo/Product Walkthrough

VisionAssist

See the world. Hear what matters.

1 Capture / Input

Smart glasses capture real-time visuals

Camera Active

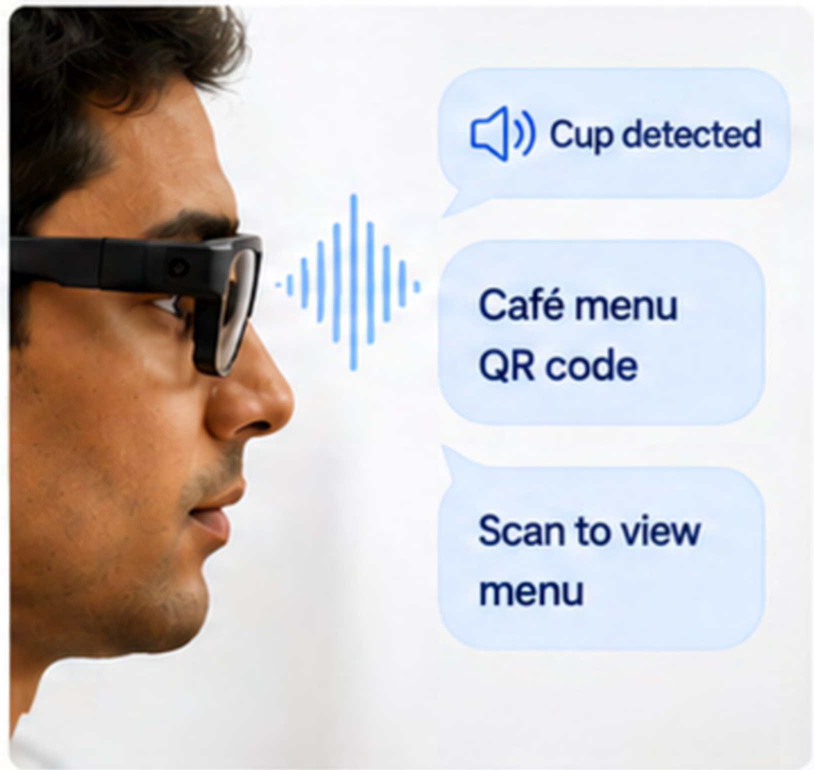
2 AI Processing

AI detects objects, reads text and understands the scene

Processing...
Object Detection + OCR

3 Voice Guidance / Output

Smart glasses provides audio feedback and reads out text



Speaking...

4 User Interaction

User takes action based on the guidance

Action Completed

Real-time Assistance | Audio Feedback | Text Recognition (OCR) | Object Detection

TECH STACK & ARCHITECTURE

Powering **VisionAssist** – Real-time AI for Visually Impaired Assistance

TECH STACK

COMPUTER VISION
- OpenCV
- YOLOv8

Real-time object detection and localization

OCR
- EasyOCR

Extracts and reads text from environment

VOICE AI
- Vosk
- pyttsx3

Enables offline voice commands and natural speech output

AI ORCHESTRATION
- OpenClaw Runtime
- Gemini API

Understands context and generates intelligent responses

CORE LANGUAGE
- Python

Powers the entire system with performance and flexibility

SYSTEM ARCHITECTURE

Camera Input — Captures real-time video from the environment

↓

YOLO Detection — Detects objects and estimates their positions (left/center/right)

↓

OCR Extraction — Reads text from signs, labels, screens, and more

↓

Scene Builder (JSON) — Combines objects, text and positions into structured JSON

↓

OpenClaw + Gemini API — Understands the scene and generates intelligent responses

↓

Spatial Audio + TTS — Converts response into direction-aware voice guidance

↓

User (Visually Impaired) — Receives real-time audio guidance to understand surroundings

KEY HIGHLIGHTS

OFFLINE FIRST
Works without internet

REAL-TIME
Instant detection and response

SPATIAL AUDIO
Direction-aware voice guidance

CROSS-PLATFORM
macOS & Windows support

PRIVACY FOCUSED
All processing on-device

Impact & Use cases

🌍 **Impact**

- Over **2.2 billion** people globally experience vision impairment (**WHO**).
- Helps visually impaired users navigate independently.
- Reduces dependency through real-time voice assistance.
- Improves safety using obstacle **detection** and **navigation** guidance.
- Enables independent reading of menus, signboards, and notices using OCR.

💡 **Real-World Use Cases**

- 🛣️ Walking on roads while receiving voice alerts about vehicles, poles, stairs, or obstacles
- 🍽️ Reading restaurant menus and shop signboards through OCR-based voice output
- 🚌 Identifying bus numbers, classroom notices, and public information boards independently
- 🏫 Assisting students in reading printed study materials and instructions
- 🚨 Helping users move safely in unfamiliar places during emergencies or crowded situations

Differentiation

Why VisionLink Stands Out

- Hands-free smart glasses experience for visually impaired users
- **Real-time scene detection** with instant voice guidance
- Combines OCR, navigation, and obstacle detection in one system
- Works even in low or no internet environments
- More affordable compared to existing smart assistive glasses

Obstacle Detection

Signboard Reading

Independence & Confidence

Scene Detection

Voice Assistance

Hands-Free Experience

✨ Our MOAT

VisionLink combines affordability, real-time scene understanding, and wearable hands-free accessibility into one unified assistive solution for visually impaired individuals.

Idea

Github Repo:

https://github.com/Soumya09-source/VisionLink.git

Demo Link:

https://drive.google.com/drive/folders/1nIcToTy2jeuUzIIx5nCc80H5_8rfQqeV?usp=sharing

Document submitted: AI disclosure

https://drive.google.com/file/d/1c73KwdUWTCt5a7cCu6rAWcTFsQr2qCVp/view?usp=sharing

Future Scope / Advanced Features:

📍 Outdoor Navigation Assistance

- Real-time GPS-based navigation
- Voice-guided outdoor directions
- Nearby landmark awareness
- Safe route assistance for visually impaired users

🧠 Visual Memory System

- Remembers important locations/objects
- User can say:
  "Remember this place"
  "Remember my bag"
- AI recalls saved locations later
- Helps users relocate important items